Data-Knowledge-Driven Self-Organizing Fuzzy Neural Network

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***Abstract*— Fuzzy neural networks (FNNs) hold the advantages of knowledge leveraging and adaptive learning, which have been widely used in nonlinear system modeling. However, it is difficult for FNNs to obtain the appropriate structure in the situation of insufficient data, which limits its generalization performance. To solve this problem, a data-knowledge-driven self-organizing FNN (DK-SOFNN) with a structure compensation strategy and a parameter reinforcement mechanism is proposed in this article. First, a structure compensation strategy is proposed to mine structural information from empirical knowledge to learn the structure of DK-SOFNN. Then, a complete model structure can be acquired by sufficient structural information. Second, a parameter reinforcement mechanism is developed to determine the parameter evolution direction of DK-SOFNN that is most suitable for the current model structure. Then, a robust model can be obtained by the interaction between parameters and dynamic structure. Finally, the proposed DK-SOFNN is theoretically analyzed on the fixed structure case and dynamic structure case. Then, the convergence conditions can be obtained to guide practical applications. The merits of DK-SOFNN are demonstrated by some benchmark problems and industrial applications.**

***Index Terms*— Fuzzy neural network (FNN), generalization performance, parameter reinforcement mechanism, structure compensation strategy.**

I. Introduction

MACHINE learning technologies mine new knowledge and skills from data to simulate or implement human behavior and have been widely used to solve complex problems in engineering applications and scientific fields [1], [2]. The research directions of machine learning include stochastic forest [3], decision tree [4], Bayesian learning [5], artificial neural network [6], [7], fuzzy system [8], [9], and so on. As one of the research hotspots of machine learning, fuzzy neural network (FNN), having the adaptive ability of neural network and the robust learning ability of fuzzy system, has been widely used in nonlinear system modeling [10], [11] and controller design [12], [13]. For the sake of improving the accuracy and generalization performance, a large number of parameter learning methods [14], [15] and structure learning methods [16], [17] have been proposed to train FNN. However, most of these methods [14]–[17] assumed that there are enough data in the learning process. If sufficient data cannot be acquired, the trained FNN will have poor generalization performance.

To enhance the generalization performance of FNN, some transfer learning methods are proposed to leverage the structure of related FNN to determine the structure of current FNN [18], [19]. For example, Zuo *et al.* [20] proposed a fuzzy transfer learning method based on the Gaussian mixture model (GM-FTL). This method can mine the data structures from the source model and target model, and then adjust the number of fuzzy rules of target model according to data structures. The experimental illustrated that this method can obtain a complete structure to improve generalization performance. In [21], a structured domain adaption method was proposed to adjust the number of fuzzy rules of FNN. This method used the instance knowledge of the source and target domains to construct the iterative structure matrix, so as to determine the structures of target FNN. It can be shown that this method can not only obtain a complete target model but also unify the structures of source FNN and target FNN. Moreover, Wang *et al.* [22] proposed a robust feature learning method, which used group Lasso penalty to mine the shared features of the source model and target model. Then, the structure of the target FNN can be determined by the shared features. These methods [18]–[22] can share the structures between the source and target models to improve the generalization performance. However, due to different complexities of the source task and target task, the structure sizes required by source FNN and target FNN are different. The unwanted structures in source FNN will reduce the accuracy of the target FNN.

To acquire an effective structure, some self-organizing methods are proposed to adjust the structure of target FNN [23], [24]. For instance, Han and Qiao [25] designed a self-organizing FNN using the index of rule sensitivity (RS-SOFNN). The index of rule sensitivity can evaluate the matching degree of the margin distribution of fuzzy rules to learning task. Then, the fuzzy rules with poor margin distribution can be deleted to acquire a compact FNN structure. In [26], a self-organizing mechanism with the index of contribution degree was proposed to adjust the structure of

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FNN. In this method, the contributions of fuzzy rules were used to evaluate their matching degree of conditional distribution to learning task. Then, useless fuzzy rules were deleted to enhance the generalization ability of FNN. In addition, Zheng *et al.* [27] leveraged the index of rule redundancy to design a self-organizing mechanism. This mechanism used spike-timing-dependent plasticity to evaluate the redundancy of fuzzy rules to adjust the FNN structure. However, most of these methods [23]–[27] only consider an aspect of index of fuzzy rules to adjust the structure of FNN. To comprehensively consider the multiple characteristics of fuzzy rules, a cooperative structure self-organization method was proposed in [28]. In this method, the indexes of contribution and redundancy of fuzzy rules were considered to adjust the structure of FNN. The results demonstrated that this method can obtain the fuzzy rules with high contribution and low redundancy. Meanwhile, a multiobjective particle swarm optimization algorithm was used to adjust the number of fuzzy rules in [29]. In this method, the indexes of rule complexity and feature complexity were considered to adjust the structure of FNN. These methods [23]–[29] can remove unwanted fuzzy rules of the FNN using the structure information of the target domain. However, due to insufficient information in the target domain, only using target information to adjust the structure cannot obtain satisfactory performance [30], [31]. Meanwhile, the above methods only focus on the adjustment of model structure and ignore the adjustment of parameters in the learning process.

In fact, the parameter learning process and structure learning process of FNN complement each other. A suitable model structure can improve the efficiency and precision of parameter learning, and a high-precision model parameter can calculate the more reliable indexes for structure adjustment [32], [33]. To learn the structure and parameters of FNN, Subrahmanya and Shin proposed a constructive method in [34]. This method learned the parameters using the particle swarm optimization algorithm and adjusted the structure using the covariance matrix adaptation strategy to improve the learning performance. In [35], the network size and the parameters of FNN were optimized using the adaptive particle swarm optimization method (APSO-FNN). Then, the compact structure and accurate parameters of FNN can be obtained to improve the learning performance. Moreover, El-Sousy [36] developed a recurrent FNN with self-evolving mechanism. This mechanism adjusted the structure and parameters of FNN at the same time. The simulation results confirmed that the proposed method can obtain robust learning performance. However, since the dynamic change in structure will influence the parameter, the parameter of growing neurons is randomly initialized and the parameter of deleted neurons is reset, which cause the uncertainty in the parameter learning process. Parameter learning cannot quickly and smoothly track the change in structure, which reduces the learning efficiency [37]–[39].

In previous works, we have designed the interactive transfer learning algorithms to adjust the structure and parameter of FNN [40]. This method adjusted the structure of target FNN using history knowledge, i.e., the structural risks of the source and target FNNs and learn the parameters using the gradient descent algorithm. However, this method has three problems: 1) this method directly assigns the structure of the target FNN to the source FNN, which ignores the difference in task complexities between the source scene and the target scene; 2) this method ignores the dynamic interaction between structure change and parameter learning and the robustness is not satisfied; and 3) there is no theoretical analysis of structure and parameter learning in this method.

In this article, a data-knowledge-driven self-organizing FNN (DK-SOFNN) is designed to learn the structure and parameters. The advantages of this DK-SOFNN lie in the following three folds.

1) A data-knowledge-driven framework is proposed to learn FNN. This framework can leverage the knowledge from related FNN to learn the structure and parameters of current FNN. Then, a complete model structure and accurate model parameters can be acquired by sufficient knowledge and data information.

2) A structure compensation strategy is proposed to learn the structure of DK-SOFNN. This strategy can mine the structure information, i.e., rule contribution, similarity, and sensitivity from related FNN to decide the structure of current FNN. Instead of directly sharing the structure of the source and target models, this strategy can acquire a complete model structure without negative transfer.

3) A parameter reinforcement mechanism is proposed to learn the parameter of DK-SOFNN. This mechanism can generate several parameter evolution directions by adjusting the knowledge-driven degree, and then select the direction which is the most suitable for the current model structure. Through the interaction between parameters and dynamic structure, the parameters can be learned smoothly.

4) The proposed DK-SOFNN is theoretically analyzed on the fixed structure case and dynamic structure case. Then, the convergence conditions can be obtained to guide practical applications.

The article is organized as follows. The basic conception of the self-organizing FNN and transfer learning is provided in Section II. The design process of DK-SOFNN and its theoretical analysis are illustrated in Sections III and IV, respectively. Finally, the results and conclusion are given in Sections V and VI, respectively. The basic notations are shown in Table I.

II. Preliminaries

In the preliminaries, the general structure of the self-organizing FNN is introduced in Section II-A, and the definition of the fuzzy transfer learning algorithm is introduced in Section II-B.

A. Self-Organizing FNN

For the structure of the traditional FNN, there are five layers, including the input layer, the radial basis function (RBF) layer, the rule layer, the normalized layer, and the output layer. The general structure of FNN can be seen in Fig. 1.

1) Input Layer: The input layer puts the input data into the FNN, which can be expressed as

$$\mathbf{x}(t) = \left[x^1(t), \ldots, x^p(t), \ldots, x^P(t)\right]^{\mathrm{T}} \tag{1}$$

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TABLE I
BASIC NOTATIONS OF DK-SOFNN

Notations	Mathematical Meanings
$\mathbf{x}(t)$	The input vector of FNN.
$\varphi(t)$	The output value of RBF neuron.
$\mathbf{c}(t)$	The vector of centers of RBF neuron.
$\boldsymbol{\sigma}(t)$	The vector of widths of RBF neuron.
$u(t)$	The output value of rule neuron.
$\mathbf{v}(t)$	The output vector of normalized layer.
$\mathbf{w}(t)$	The vector of connecting weight.
D	The domain.
F	The features.
$P(X)$	The data distribution.
T	The task.
Y	The label.
$f(\cdot)$	The FNN.
$E_S(n)$	The least square criterion function of source FNN.
$\boldsymbol{\Omega}_S(n)$	The parameters of source FNN.
$R_S^l(n)$	The similarity index of lth source fuzzy rule.
$M_S^l(n)$	The sensitivity index of lth source fuzzy rule.
$C_S^l(n)$	The contribution index of lth source fuzzy rule.
$\bar{R}_T(n)$	The average value of similarity of target fuzzy rules.
$\bar{R}_S(n)$	The average value of similarity of source fuzzy rules .
$\bar{M}_T(n)$	The average value of sensitivity of target fuzzy rules.
$\bar{M}_S(n)$	The average value of sensitivity of source fuzzy rules.
$\bar{C}_T(n)$	The average value of contribution of target fuzzy rules.
$\bar{C}_S(n)$	The average value of contribution of source fuzzy rules.
$E_T^h(t)$	The hth least square criterion function of target FNN.
$\alpha^h(t)$	The hth weight of data driven part.
$\beta^h(t)$	The hth weight of knowledge driven part.
$\boldsymbol{\Omega}_T(t)$	The vector of the parameters of target FNN.
$\Delta^c(t)$	The short term of center.
$\Delta^\sigma(t)$	The short term of width.
$W^h(t)$	The evaluation weight learned by the hth framework.
$e_T^h(t+n)$	The output error of target FNN using $\boldsymbol{\Omega}_T^h(t+1)$.

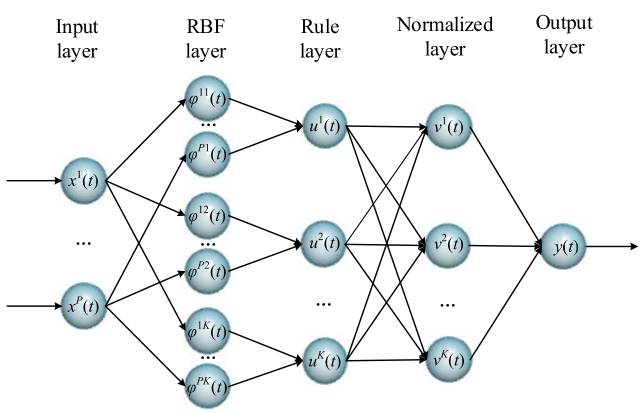

Fig. 1. Structure of FNN.

where $\mathbf{x}(t)$ is the input vector, and $x^p(t)$ is the input of the pth input neuron, $p = 1, \ldots, P$.

2) RBF Layer: The RBF layer fuzzifies the input data by mapping the input space into the Gaussian distribution space and calculates the fuzzy membership degree of the input data

$$\varphi^{pk}(t) = \exp\left(-\frac{\left(x^p(t) - c^{pk}(t)\right)^2}{2\left(\sigma^{pk}(t)\right)^2}\right) \tag{2}$$

where $\varphi^{pk}(t)$ is the fuzzy membership degree of the pth input data to the kth fuzzy rule, $k = 1, \ldots, K$, and $c^{pk}(t)$ and $\sigma^{pk}(t)$ are the pth center and the width of the kth RBF neuron, respectively.

3) Rule Layer: The rule layer represents the data as fuzzy rules and calculates the total fuzzy membership degree of the data

$$u^k(t) = \prod_{p=1}^{P} \exp\left(-\frac{\left(x^p(t) - c^{pk}(t)\right)^2}{2\left(\sigma^{pk}(t)\right)^2}\right) \tag{3}$$

where $u^k(t)$ is the output value of the kth rule neuron.

4) Normalized Layer: The generalized layer computes the output of each fuzzy rule

$$v^k(t) = \frac{u^k(t)}{\sum_{k=1}^{K} u^k(t)} \tag{4}$$

where $v^k(t)$ is the output value of the kth generalized neuron.

5) Output Layer: The output layer integrates the outputs of fuzzy rules

$$y(t) = \sum_{k=1}^{K} w^k(t) v^k(t) \tag{5}$$

where $y(t)$ is the output of FNN, and $w^k(t)$ is the weight of the kth generalized neuron.

The structure of FNN determines the generalization performance and computational complexity. To obtain a compact FNN structure, many self-organizing algorithms are proposed for FNN. These methods have the following steps: First, the indexes of fuzzy rules are calculated to evaluate the structure. Second, the structure is enlarged or shrunk according to the indexes of fuzzy rules. Finally, the fuzzy rules are adjusted to reduce the error changes caused by structure change.

The self-organizing FNNs can be divided into three types of increment, decrement, and elasticity. The incremental self-organizing FNN initializes a small network structure and selects the most effective fuzzy rule according to the evaluation indexes. Then, the effective fuzzy rules are divided into multiple neurons to realize the incremental adjustment of the network structure. Because the incremental self-organizing FNN sets the lower limit of the network structure, the network precision can be improved on the basis of good generalization performance. The decremented self-organizing FNN initializes a large network structure. Then, the ineffective fuzzy rules are deleted according to the evaluation index to realize the decremented adjustment of the network structure. Because the decremented self-organizing FNN sets the upper limit of the network structure, the generalization performance can be improved on the basis of good network accuracy. To balance the network accuracy and generalization performance, the elastic self-organizing FNN (in Fig. 2) is proposed to grow and prune neurons simultaneously, where the neurons connected by the red line are the pruned neuron and the neurons connected by the blue line are the growing neurons, and $\varphi^{pZ}(t)$ is the fuzzy membership degree of the pth input data to the Zth fuzzy rule.

FNN has the advantage of knowledge leveraging and adaptive learning, which has been widely used in modeling nonlinear system. However, FNNs are difficult to obtain appropriate

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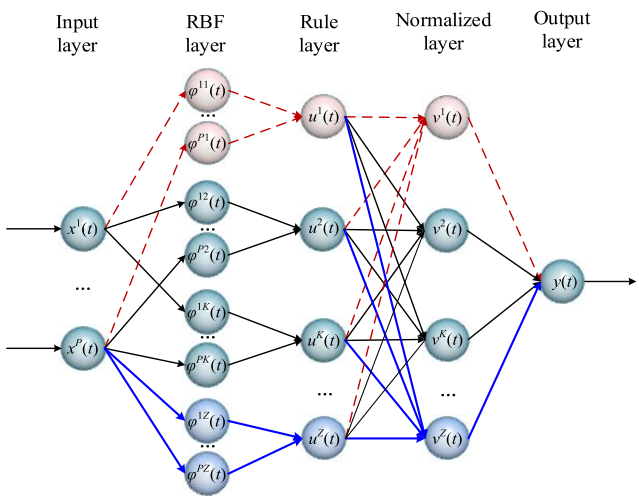

Fig. 2. Self-organizing FNN.

structure in the situation of insufficient data, which limits its generalization performance. Therefore, the fuzzy transfer learning algorithm is proposed to learn FNN using knowledge and data.

B. Fuzzy Transfer Learning Algorithm

The fuzzy transfer learning algorithm used history knowledge from the source domain $D_S = \{F_S,\ P_S(X)\}$ and the source task $T_S = \{Y_S,\ f_S(\cdot)\}$ to improve the performance of the target FNN $f_T(\cdot)$, where F_S is the source feature, $P_S(X)$ is the source data distribution, Y_S is the source label, and $f_S(\cdot)$ is the source FNN. This article focused on inductive fuzzy transfer learning where the target task is different from the source task, no matter when the source and target domains are the same or not. The general framework of the inductive fuzzy transfer learning algorithm can be given as

$$\begin{aligned} &\arg\min_{f_T(t)} \sum_{t=1}^{N_T} l(f_T(\mathbf{x}_T(t), \mathbf{K}_S(t)), y_T(t)) \\ &\text{s.t. } f_T(t) \neq f_S(t) \end{aligned} \tag{6}$$

where $\{\mathbf{x}_T(t),\ y_T(t)\}$ is the instance of target domain, $t = 1, \ldots, N_T$, N_T is the number of target instance, $\mathbf{K}_S(t)$ is the knowledge extracted from source instance $\{\mathbf{x}_S(n),\ y_S(n)\}$, $n = 1, \ldots, N_S$, and N_S is the number of source instance.

The inductive fuzzy transfer learning algorithm can process the problem of insufficient data. However, most inductive fuzzy transfer learning algorithms only focus on the learning of parameters and ignore the learning of structures. In fact, the parameter learning process and structure learning process of FNN complement each other. Therefore, a DK-SOFNN is proposed in Section III.

III. Data-Knowledge-Driven Self-Organizing FNN

In this part, the designed DK-SOFNN is used to learn the parameters and structure of FNN using knowledge and data (in Fig. 3). The proposed DK-SOFNN contains the following three steps. First, the source FNN is learned to mine the structure knowledge. Second, a structure compensation algorithm is proposed to adjust the number of target fuzzy rule using the structure knowledge, so as to acquire the complete target FNN. Finally, a parameter reinforcement algorithm is designed to optimize the parameters of the target FNN that satisfy the current dynamic structure, so as to realize robust learning of the parameters of FNN. Then, a DK-SOFNN with preference accuracy and generation performance can be obtained.

A. Mining the Structure Knowledge From Source FNN

The source FNN, constructed by source instances, can be given as

$$y_S(n) = \sum_{k=1}^{K} \frac{w_S^k(n) \prod_{p=1}^{P} \exp\left(-\frac{\left(x_S^p(n) - c_S^{pk}(n)\right)^2}{2\left(\sigma_S^{pk}(n)\right)^2}\right)}{\sum_{k=1}^{K} \prod_{p=1}^{P} \exp\left(-\frac{\left(x_S^p(n) - c_S^{pk}(n)\right)^2}{2\left(\sigma_S^{pk}(n)\right)^2}\right)} \tag{7}$$

where $y_S(n)$, $\mathbf{x}_S(n)$ are the output and the input of source FNN, respectively, $\mathbf{w}_S(n)$, $\mathbf{c}kS(t)$, and $\sigma kS(t)$ are the weights, centers, and widths of source FNN, respectively, and P and K are the number of input neurons and RBF neurons, respectively.

The back propagation algorithm is used to learn the parameters of source FNN

$$\mathbf{\Omega}_S(n+1) = \mathbf{\Omega}_S(n) - \lambda_S \frac{\partial E_S(n)}{\partial \mathbf{\Omega}_S(n)} \tag{8}$$

where λ_S is the learning rate of source FNN, $\Omega_S(n) = [\mathbf{k}_S^1(n)^{\mathrm{T}}, \ldots, \mathbf{k}_S^K(n)^{\mathrm{T}}]^{\mathrm{T}}$ is the vector of the parameters of source FNN, $\mathbf{k}_S^k(t) = [\mathbf{c}_S^k(n)^{\mathrm{T}},\ \sigma_S^k(n)^{\mathrm{T}},\ w_S^k(n)]^{\mathrm{T}}$ are the parameters of the kth source fuzzy rule, and $E_S(n)$ is the objection function of source FNN at time n

$$E_S(n) = \frac{1}{2} e_S(n)^2 \tag{9}$$

where $e_S(n) = y_S(n) - y_{Sd}(n)$ is the output error of source FNN at time n, and $y_{Sd}(n)$ is the desired output value at time n.

It can be seen that the FNN with a larger structure has preferable accuracy but worse computational burden and generalization. On the contrary, FNN with a smaller structure has preferable computational burden and generalization but worse accuracy. To balance the accuracy and computational burden, the indexes of rule redundancy, sensitivity, and contribution are used to evaluate the effectiveness of neuron. The redundancy index is used to prune useless neuron, and rule sensitivity and contribution are used to prune useless neuron or duplicate effective neuron. In general, similarity can evaluate the redundancy of fuzzy rules, sensitivity can evaluate the matching degree of margin distribution of fuzzy rules to the learning task, and contribution can evaluate the matching degree of condition distribution of fuzzy rules to the learning task. An FNN structure with smaller redundancy and larger matching degree of margin distribution can cover wider data distribution and has better generalization performance. Meanwhile, an FNN structure with larger matching degree of conditional distribution can predict the data label with higher

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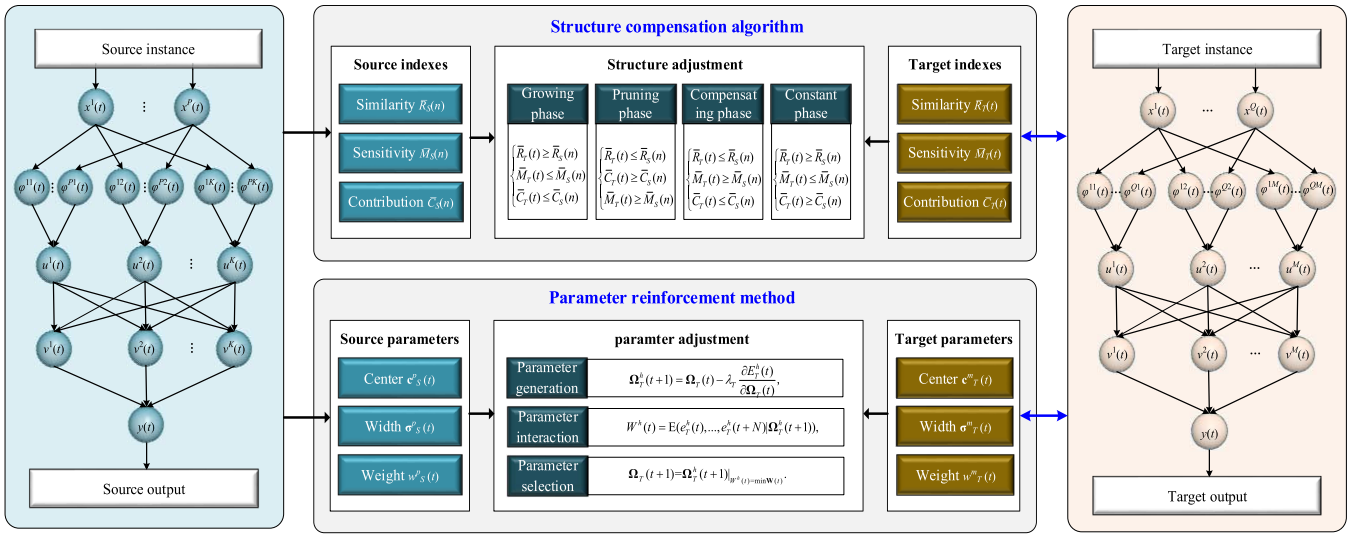

Fig. 3. DK-SOFNN.

accuracy. Therefore, rule similarity and sensitivity are used to evaluate the generalization performance of FNN structure. The rule contribution is used to evaluate the accuracy of FNN structure. The objective of source FNN is to minimize the similarity and maximize the sensitivity and contribution. The index of similarity can be defined as (10), shown at the bottom of the page, where $R_S^l(n)$ is the similarity of the lth fuzzy rule with other fuzzy rules at time n. N is the sample interval in the structure adjustment process, $u_S^l(n)$ and $u_S^k(n)$ are the output value of the lth and kth rule neuron in source FNN, respectively, and $\bar{u}_S(n)$ is the average output value of the rule neuron in source FNN. The index of sensitivity is

$$M_S^l(n) = \frac{\sum_{n}^{n-N+1} u_S^l(n)}{\sum_{n}^{n-N+1} \sum_{k=1}^{K} u_S^k(n)} \tag{11}$$

where $Ml\ S(n)$ is the sensitivity of the lth fuzzy rule at time n. The index of contribution can be defined as

$$\begin{aligned} C_S^l(n) &= 1/d_S^l(n) \\ d_S^l(n) &= \frac{\sum_{n}^{n-T+1} \sum_{k=1}^{K} \left(u_S^l(n) - y_S(n)\right)\left(u_S^k(n) - y_S(n)\right)}{\sqrt{\sum_{n}^{n-N+1} \left(u_S^l(n) - y_S(n)\right)^2} \sqrt{\sum_{n}^{n-N+1} \sum_{k=1}^{K} \left(u_S^k(n) - y_S(n)\right)^2}} \end{aligned} \tag{12}$$

where $C_S^l(n)$ is the contribution of the lth fuzzy rule at time n. Based on the indexes, the growth and pruning criterions of source FNN are designed to adjust the structure of FNN. The growing criterion is activated if the lth fuzzy rule satisfies the follow condition:

$$\begin{cases} R_S^l(n) = \min \mathbf{S}_S(n) \\ M_S^l(n) = \max \mathbf{M}_S(n) \\ C_S^l(n) = \max \mathbf{C}_S(n) \end{cases} \tag{13}$$

where $\mathbf{R}_S(n) = [R1\ S(n), \ldots, RK\ S(n)]$ is the similarity vector of source fuzzy rules at time n, $\mathbf{M}_S(n) = [M_S^1(n), \ldots, M_S^K(n)]$ is the sensitivity vector of source fuzzy rules at time n, and $\mathbf{C}_S(n) = [C_S^1(n), \ldots, C_S^K(n)]$ is the contribution vector of source fuzzy rules at time n. Then, the initial parameters of new fuzzy rules are given as

$$\begin{cases} c_S^{p\text{new}}(n) = x_S^p(n) \\ \sigma_S^{p\text{new}}(n) = \left|x_S^p(n) - c_S^{pl}(n)\right| \big/ 2 \\ w_S^{\text{new}}(n) = w_S^l(n) \end{cases} \tag{14}$$

where $c_S^{p\text{new}}(n)$ is the pth center value of the new fuzzy rule at time n, $x_S^p(n)$ is the pth input at time n, $\sigma_S^{p\text{new}}(n)$ is the pth width value of the new fuzzy rule at time n, $c_S^{pl}(n)$ is the pth center value of the lth source fuzzy rule at time n, $w_S^{\text{new}}(n)$ is the weight of the new source fuzzy rule at time n, and $w_S^l(n)$ is the weight of the lth source fuzzy rule at time n. The pruning criterion is activated if the indexes of the lth

$$R_S^l(n) = 1/\frac{\sum_{n}^{n-N+1} \sum_{k=1}^{K} \left(u_S^l(n) - \bar{u}_S(n)\right)\left(u_S^k(n) - \bar{u}_S(n)\right)}{\sqrt{\sum_{n}^{n-N+1} \left(u_S^l(n) - \bar{u}_S(n)\right)^2} \sqrt{\sum_{n}^{n-N+1} \sum_{k=1}^{K} \left(u_S^k(n) - \bar{u}_S(n)\right)^2}} \tag{10}$$

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fuzzy rule satisfy

$$\begin{cases} R_S^l(n) = \max \mathbf{S}_S(n) \\ M_S^l(n) = \min \mathbf{M}_S(n) \\ C_S^l(n) = \min \mathbf{C}_S(n). \end{cases} \tag{15}$$

Then, the parameters of the lth fuzzy rule are reset to zero to prune useless neuron.

In the source domain, there are sufficient samples to train the source FNN. Therefore, a source FNN with high precision and strong generalization performance can be obtained by parameter learning and structure learning. However, the data in the target domain are insufficient to obtain a complete target FNN. For the purpose of balancing precision and generalization performance of the target FNN, the structure information (rule similarity, sensitivity, and contribution) of source FNN can be used to adjust the fuzzy rule numbers of target FNN.

B. Adjust the Structure of Target FNN Using Structure Compensation Algorithm

The structure of the target FNN is given as

$$y_T(t) = \sum_{b=1}^{B} \frac{w_T^b(t) \prod_{q=1}^{Q} \exp\left(-\frac{\left(x_T^q(t) - c_T^{qb}(t)\right)^2}{2\left(\sigma_T^{qb}(t)\right)^2}\right)}{\sum_{b=1}^{B} \prod_{q=1}^{Q} \exp\left(-\frac{\left(x_T^q(t) - c_T^{qb}(t)\right)^2}{2\left(\sigma_T^{qb}(t)\right)^2}\right)} \tag{16}$$

where $y_T(t)$ and $\mathbf{x}_T(t)$ are the output and input of the target FNN, respectively, Q is the number of input neurons, $\mathbf{w}_T(t)$, $\mathbf{c}_T^b(t)$, and $\boldsymbol{\sigma}_T^b(t)$ are the vector of weights, centers, and widths of the target FNN, respectively, and M and Q are the number of RBF neurons and the input neurons, respectively.

To learn the structure of the target FNN with insufficient data, a structure compensation algorithm is proposed for target FNN using the structure knowledge of the source FNN. The proposed structure compensation algorithm can evaluate the structure of the target FNN using the structure knowledge of the source FNN. According to the evaluation results, the structure of the target FNN is adjusted and compensated using the structure of the source FNN. The adjustment process of the target FNN is divided into growing phase, pruning phase, compensating phase, and constant phase.

1) Growing Phase: If the structure of the target FNN has a lower similarity, higher sensitivity and lower contribution than that of the source FNN, it indicates that the generalization performance of the target FNN is strong but the accuracy of the target FNN is poor. In general, a large target FNN structure usually has poor generalization performance but high accuracy. Therefore, the structure of the target FNN needs to be extended to improve the accuracy. The growing condition of the target FNN can be given as

$$\begin{cases} \bar{R}_T(t) \le \bar{R}_S(n) \\ \bar{M}_T(t) \ge \bar{M}_S(n) \\ \bar{C}_T(t) \le \bar{C}_S(n) \end{cases} \tag{17}$$

where $\bar{R}_T(t)$ is the average value of similarity of target fuzzy rules at time n, $\bar{R}_S(n)$ is the average value of similarity of source fuzzy rules at time n, $\bar{M}_T(t)$ is the average value of sensitivity of target fuzzy rules at time t, $\bar{M}_S(n)$ is the average value of sensitivity of source fuzzy rules at time n, $\bar{S}_T(t)$ is the average value of contribution of target fuzzy rules at time t, and $\bar{S}_S(n)$ is the average value of contribution of source fuzzy rules at time n.

To improve the accuracy of target FNN, the target fuzzy rule with largest contribution is growing in the growing phase. Assuming that the lth fuzzy rule has the largest contribution, the initial parameters of new fuzzy rule are initialed as

$$\begin{cases} c_T^{q\text{new}}(t) = x_T^q(t) \\ \sigma_T^{q\text{new}}(t) = \left| x_T^q(t) - c_T^{ql}(t) \right| / 2 \\ w_T^{\text{new}}(t) = w_T^l(t) \end{cases} \tag{18}$$

where $c_T^{q\text{new}}(t)$ is the qth center value of the new fuzzy rule at time t, $x_T^q(t)$ is the qth target input at time t, $\sigma_T^{q\text{new}}(t)$ is the qth width value of the new fuzzy rule at time t, $c_T^{ql}(t)$ is the qth center value of the lth target fuzzy rule at time t, $w_T^{\text{new}}(t)$ is the weight of the new fuzzy rule at time t, and $w_T^l(t)$ is the weight of the lth target fuzzy rule at time t.

2) Pruning Phase: If the structure of the target FNN has a higher contribution and lower sensitivity or higher similarity than that of the source FNN, the generalization performance of the target FNN is poor but the accuracy of the target FNN is high. In general, a small target FNN structure has strong generalization performance and low accuracy. Therefore, the structure of the target FNN needs to be pruned to improve generalization performance. The pruning condition of the target FNN can be given as

$$\begin{cases} \bar{R}_T(t) \ge \bar{R}_S(n) \\ \bar{C}_T(t) \ge \bar{C}_S(n) \end{cases} \quad \text{or} \quad \begin{cases} \bar{M}_T(t) \le \bar{M}_S(n) \\ \bar{C}_T(t) \ge \bar{C}_S(n). \end{cases} \tag{19}$$

To improve the generalization performance of the target FNN, the target fuzzy rule with lowest sensitivity or highest similarity is pruned in the pruning phase. Based on the characteristics of sensitivity and similarity, the pruning phase can be divided into the following three situations: 1) if the structure of the target FNN has a lower sensitivity and lower similarity than the source FNN, the target fuzzy rule with smallest sensitivity is pruned in the pruning phase; 2) if the structure of the target FNN has a higher sensitivity and higher similarity than the source FNN, the target fuzzy rule with highest similarity is pruned in the pruning phase; and 3) if the structure of the target FNN has a low sensitivity and high similarity than the source FNN, select the worst one from the two target fuzzy rules to be pruned in the pruning phase.

3) Compensating Phase: If the structure of the target FNN has a lower sensitivity or higher similarity and lower contribution than that of the source FNN, both the generalization performance and the accuracy of the target FNN are poor. Therefore, the structure of the target FNN needs to be compensated to improve the generalization performance and accuracy. The compensating condition of the target FNN can be given as

$$\begin{cases} \bar{R}_T(t) \ge \bar{R}_S(n) \\ \bar{C}_T(t) \le \bar{C}_S(n) \end{cases} \quad \text{or} \quad \begin{cases} \bar{M}_T(t) \le \bar{M}_S(n) \\ \bar{C}_T(t) \le \bar{C}_S(n). \end{cases} \tag{20}$$

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To improve the performance of the target FNN, the target fuzzy rule with the worst performance is replaced by the source fuzzy rule with the best performance, where the performance is evaluated by the sum of three indexes. Assuming that the lth target fuzzy rule has the worst performance and the zth source fuzzy rule has the best performance, the reset parameters of the lth target fuzzy rule are

$$\begin{cases} \mathbf{c}_T^l(t) = \mathbf{c}_S^z(n) \\ \boldsymbol{\sigma}_T^l(t) = \boldsymbol{\sigma}_S^z(n) \\ w_T^l(t) = w_S^z(n) \end{cases} \tag{21}$$

where $\mathbf{c}_T^l(t)$ is the lth target fuzzy rule at time t, $\mathbf{c}_S^z(n)$ is the zth source fuzzy rule at time n, $\boldsymbol{\sigma}_T^l(t)$ is the lth target fuzzy rule at time t, $\boldsymbol{\sigma}_S^z(n)$ is the zth source fuzzy rule at time n, $w_T^l(t)$ is the lth target fuzzy rule at time t, and $w_S^z(n)$ is the zth source fuzzy rule at time n.

4) Constant Phase: If the structure of the target FNN has a higher sensitivity, lower similarity, and higher contribution than the source FNN, the target FNN has strong generalization performance and high accuracy. Therefore, the structure of the target FNN does not need to be adjusted. The constant condition of the target FNN can be given as

$$\begin{cases} \bar{R}_T(t) \geq \bar{R}_S(n) \\ \bar{M}_T(t) \leq \bar{M}_S(n) \\ \bar{C}_T(t) \geq \bar{C}_S(n). \end{cases} \tag{22}$$

Remark 1: In the structure compensation algorithm, the structure information of the source FNN can be leveraged to adjust the structure of the target FNN in (14)–(17). In (18), the structure of the target FNN can be compensated by the source FNN. Moreover, the indexes of rule similarity, sensitivity, and contribution defined in (9) and (10) are used to balance the generalization performance and accuracy.

C. Optimize the Parameters of Target FNN Using Parameter Reinforcement Algorithm

The dynamic structure change in target FNN will increase the uncertainty of the training process. To optimize the parameters of the target FNN under dynamic structural changes, a parameter reinforcement method is proposed to improve the robustness of the learning process. The proposed parameter reinforcement method can learn the knowledge of the source domain and the data of the target domain to generate several groups of parameters of the target FNN. Then, the value function is designed to select the optimal parameters under the current structure.

To generate several groups of parameters of target FNN, there are H learning frameworks built for the target FNN. The hth learning framework built for the target FNN can be given as

$$E_T^h(t) = \alpha^h(t)(y_T(t) - y_{Td}(t))^2 + \beta^h(t)(\boldsymbol{\Omega}_T(t) - \mathbf{K}_S(n))^2 \tag{23}$$

where $E_T^h(t)$ is the hth least square criterion function of the target FNN, $h = 1,\ldots, H$, and $y_{Td}(t)$ is the desired output value of the target FNN at time t. $\alpha^h(t)$ is the hth data-driven weight, $\beta^h(t)$ is the hth knowledge-driven weight, $\boldsymbol{\Omega}_T(t) = [\mathbf{k}_T^1(t)^{\mathrm{T}}, \ldots, \mathbf{k}_T^B(t)^{\mathrm{T}}]^{\mathrm{T}}$ is the vector of the parameters of target FNN, $\mathbf{K}_S(n) = [\mathbf{k}_S^1(n)^{\mathrm{T}}, \ldots, \mathbf{k}_S^K(n)^{\mathrm{T}}]^{\mathrm{T}}$ is the knowledge of the source FNN, and $\mathbf{k}_T^b(t) = [\mathbf{c}_T^b(t)^{\mathrm{T}}, \boldsymbol{\sigma}_T^b(t)^{\mathrm{T}}, w_T^b(t)]^{\mathrm{T}}$ are the parameters of the bth target fuzzy rule. Then, the back propagation algorithm is used to update the parameters of target FNN

$$\boldsymbol{\Omega}_T^h(t+1) = \boldsymbol{\Omega}_T(t) - \lambda_T \frac{\partial E_T^h(t)}{\partial \boldsymbol{\Omega}_T(t)} \tag{24}$$

where $\Omega_T^h(t+1)$ is the parameters of the target FNN learned by the hth framework, λ_T is the learning rate of the target FNN, and $\partial E_T^h(t)/\partial \Omega_T(t)$ is the partial derivative of the target least square criterion function for the parameters of target FNN

$$\frac{\partial E_T^h(t)}{\partial \boldsymbol{\Omega}_T(t)} = \left[\frac{\partial E_T^h(t)}{\partial \mathbf{k}_T^1(t)}^{\mathrm{T}}, \ldots, \frac{\partial E_T^h(t)}{\partial \mathbf{k}_T^B(t)}^{\mathrm{T}}\right]^{\mathrm{T}} \tag{25}$$

where $\partial E_T^h(t)/\partial \mathbf{k}_T^b(t)$ is the partial derivative of the target least square criterion function for the parameters of the bth target fuzzy rule

$$\frac{\partial E_T^h(t)}{\partial \mathbf{k}_T^b(t)} = \left[\frac{\partial E_T^h(t)}{\partial \mathbf{c}_T^b(t)}^{\mathrm{T}}, \frac{\partial E_T^h(t)}{\partial \boldsymbol{\sigma}_T^b(t)}^{\mathrm{T}}, \frac{\partial E_T^h(t)}{\partial w_T^b(t)}\right]^{\mathrm{T}} \tag{26}$$

and

$$\begin{cases} \dfrac{\partial E_T^h(t)}{\partial \mathbf{c}_T^b(t)} = \alpha^h(t) - 2e_T(t)\Delta^c(t) + \beta^h(t)(\mathbf{c}_T^b(t) - \mathbf{c}_S^b(t)) \\ \dfrac{\partial E_T^h(t)}{\partial \boldsymbol{\sigma}_T^b(t)} = \alpha^h(t) - 2e_T(t)\Delta^\sigma(t) + \beta^h(t)(\boldsymbol{\sigma}_T^b(t) - \boldsymbol{\sigma}_S^b(t)) \\ \dfrac{\partial E_T^h(t)}{\partial w_T^b(t)} = \alpha^h(t)e_T(t) \cdot v_T^b(t) + \beta^h(t)(w_T^b(t) - w_S^b(t)) \end{cases} \tag{27}$$

where $e_T(t) = y_T(t) - y_{Td}(t)$ is the output error of the target FNN at time n, $\Delta^c(t) = w_T^b(t)v_T^b(t)(\mathbf{x}_T(t) - \mathbf{c}_T^b(t))/\ \sigma_T^b(t)^2$ is the short term of center, and $\Delta^\sigma(t) = w_T^b(t)v_T^b(t)(\mathbf{x}_T(t) - \mathbf{c}_T^b(t))^2/\sigma_T^b(t)^2$ is the short term of width. Then, several groups of parameters of target FNN can be acquired with different data-driven weight and knowledge-driven weight. Then the value function is designed to evaluate these parameters. The value function contains the error from time t to time $t + N$, which can evaluate the stability of current parameter

$$W^h(t) = E\left(e_T^h(t), \ldots, e_T^h(t+N) | \boldsymbol{\Omega}_T^h(t+1)\right) \tag{28}$$

where $W^h(t)$ is the evaluation weight of the parameters learned by the hth framework, $\mathbf{W}(t) = [W^1(t),\ldots, W^H(t)]$ is the vector of evaluation weight, and $e_T^h(t+n)$ is the output error of target FNN using $\Omega_T^h(t+1)$ at time $t+n$

$$e_T^h(t+n) = y_T\left(\boldsymbol{\Omega}_T^h(t+1), \mathbf{x}(t+n)\right) - y_{Td}(t+n). \tag{29}$$

Then, the optimal parameters under the current structure can be selected using value function

$$\boldsymbol{\Omega}_T(t+1) = \boldsymbol{\Omega}_T^h(t+1)|_{W^h(t)=\min \mathbf{W}(t)}. \tag{30}$$

Remark 2: It can be seen that the parameter reinforcement method can learn the knowledge from the source domain in (19). Moreover, the parameters are interacted with structure in

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TABLE II
ALGORITHM OF DK-SOFNN

Algorithm ITL-FNN	
Input: $\mathbf{V}_S$, $\mathbf{V}_T$, $\lambda_S(t)$=0.1, $\lambda_T(t)$=0.05, I=300, N_T=50.	
Output: $\mathbf{\Omega}_T(t)$	
1. i=1	
while $i<I$	
2. Compute $\mathbf{R}_S(n)$, $\mathbf{M}_S(n)$, $\mathbf{C}_S(n)$	%Eqs. (10)-(12)
3. Grow the effective fuzzy rule	%Eqs. (13)-(14)
4. Prune the ineffective fuzzy rule	%Eq. (15)
5 Optimize the parameter of source FNN	%Eqs. (7)-(9)
while $t<N_T$	
6 Grow the fuzzy rule in growing phase	%Eqs. (17)-(18)
7 prune the fuzzy rule in growing phase	%Eq. (19)
8 Replace the fuzzy rule in compensating phase	%Eqs. (20)-(21)
9 Keep the fuzzy rule in constant phase	%Eq. (22)
10 Compute $E_T^h(t)$	%Eq. (23)
12 Generate $\mathbf{\Omega}_T^h(t+1)$	%Eqs. (24)-(28)
13 Select out $\mathbf{\Omega}_T(t+1)$	%Eqs. (29)-(30)
14. t=t+1;	
15. i=i+1;	
16. predict $y_T(t+1)$ using$\mathbf{\Omega}_T(t+1)$	
Output: Predicted label $y_T(t+1)$, $\mathbf{\Omega}_T(t+1)$.	

(23)–(27) to get the highest reward in the dynamic environment. Then, the optimal parameters under the current structure can be selected in (28)–(30).

IV. Theoretical Analysis of DF-SOFNN

In this section, the computational complexity and convergence of DK-SOFNN are theoretically analyzed.

A. Computational Complexity

Based on the training process in Table II, denote the number of iterations as Z and denote the input dimensionality as V. Then, the computational complexity of the framework consists of two parts:

1) train source FNN: $O(ZN_SV^2)$ and
2) train target FNN: $O(hZN_TV^2)$.

In summary, the computational complexity of DK-SOFNN is $O(Z(N_S+hN_T)V^2)$. Compared with the general gradient-based approaches, the proposed DK-SOFNN needs an additional $O((h-1)ZN_TV^2)$ to generate several groups of parameters.

B. Convergence Analysis

The convergence conditions under fixed phase and dynamic phase are developed, separately.

Theorem 1 (Convergence of Fixed Phase): Suppose an FNN that is trained by the parameter reinforcement method, assume that $e_F(t) = \mathbf{\Omega}_T(t) - \mathbf{K}_S(n)$, if the learning rates of target FNN satisfies

$$0 < \lambda_T < \frac{2\left(\alpha(t)e_T(t)^2 + \beta(t)e_F(t)^2\right)}{\left(e_T(t)^2\frac{\partial y_T(t)}{\partial \mathbf{\Omega}_T(t)} + e_F(t)^2\right)^2}. \tag{31}$$

Then, the convergence of DK-SOFNN can be guaranteed, and moreover one has that $\mathbf{\Omega}_T(t)$

$$\lim_{t\to\infty} E_T^h(t) = 0. \tag{32}$$

Proof Define the Lyapunov function candidate as

$$V(\mathbf{\Omega}_T(t)) = \frac{1}{2}E_T^h(t)^{\mathrm{T}}E_T^h(t). \tag{33}$$

Calculating the difference in the Lyapunov function $V(\mathbf{k}_T(t))$ gives

$$\Delta V(\mathbf{\Omega}_T(t)) = \frac{1}{2}\left(E_T^h(t+1)^2 - E_T^h(t)^2\right) \tag{34}$$

where

$$E_T^h(t+1) = E_T^h(t) + \Delta E_T^h(t) \tag{35}$$

$$\Delta E_T^h(t) = \frac{\partial E_T^h(t)}{\mathbf{\Omega}_T(t)}(\mathbf{\Omega}_T(t+1) - \mathbf{\Omega}_T(t)). \tag{36}$$

Using the updated rule in (24), it holds that

$$\mathbf{\Omega}_T(t+1) - \mathbf{\Omega}_T(t) = -\lambda_T\frac{\partial E_T^h(t)}{\partial \mathbf{\Omega}_T(t)}. \tag{37}$$

Combining (33)–(37) together yields

$$\begin{aligned}\Delta V(\mathbf{\Omega}_T(t)) &= \lambda_T(t)\left(e_T(t)\frac{\partial y(t)}{\partial \mathbf{\Omega}_T(t)} + e_F(t)\right)^2 \\ &\quad\times\left(\frac{1}{2}\lambda_T(t)\left(e_T(t)\frac{\partial y(t)}{\partial \mathbf{\Omega}_T(t)} + e_F(t)\right)^2\right. \\ &\quad\left.-\left(\alpha(t)e_T(t)^2 + \beta(t)e_F(t)^2\right)\right).\end{aligned} \tag{38}$$

If the following condition is satisfied, it can be concluded that:

$$\Delta V(\mathbf{\Omega}_T(t)) < 0. \tag{39}$$

Hence, according to the Lyapunov-like lemma, the convergence of DK-SOFNN can be guaranteed.

Theorem 2 (Convergence of Dynamic Phase): For expletive knowledge, the parameter is initialed as $\mathbf{\Omega}^*{}_T(t) = \mathbf{\Omega}_T(t) + \lambda_T\partial E_T^h(t)/\partial\mathbf{\Omega}_T(t)$. Then, the approximation error can be given as

$$\begin{aligned}e^*(t) &= \left(y\left(\mathbf{\Omega}_T^*(t), \mathbf{x}_T(t)\right) - y_T(t)\right) \\ &= 0.\end{aligned} \tag{40}$$

The approximation error does not change after adding. The learning process is smooth, and DK-SOFNN is convergent.

Remark 3: The computational complexity of the proposed parameter reinforcement method is $O(Z(N_S+hN_T)W^2)$. Moreover, Theorem 1 shows that DK-SOFNN converges with the fixed learning rate λ_T. It is clearly seen that DK-SOFNN is also convergent with the variable knowledge structure in Theorem 2. The convergence property of DK-SOFNN is the key for successful applications.

V. Experiments on Synthetic Datasets

In this section, synthetic and real-world datasets are used to evaluate the accuracy and generalization performance of the proposed DK-SOFNN.

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A. Data Preparation

The performance of DK-SOFNN is verified by synthetic datasets and real-world datasets. The synthetic datasets come from the Mackey–Glass chaotic system. The real-world datasets are selected from the University of California at Irvine, Irvine, CA, USA, Machine Learning Repository of combined cycle power plant and quantitative structure activity relationship (QSAR) fish toxicity.

1) Generation of Mackey–Glass Chaotic Datasets: The datasets of the Mackey–Glass chaotic system are used to evaluate the proposed DK-SOFNN

$$\frac{dx(t)}{dt} = 0.9x(t) + \frac{c \cdot x(t-\tau)}{1 + x^{10}(t-\tau)} \tag{41}$$

where $x(t-24)$, $x(t-18)$, $x(t-12)$, and $x(t-6)$ are the input variables, $x(t)$ is the output of such a system, and c is the scale coefficient.

For the Mackey–Glass chaotic datasets, the scale coefficient is set as $c = 0.2$ to generate the source datasets. The size of the source training sample is 900 within the interval [1, 900]. Meanwhile, $c = 0.21$ to generate the target datasets. The data within the interval [901, 1000] are chosen as the target training sample. The data within the interval [1201, 1300] are chosen as the target testing sample.

2) Generation of Combined Cycle Power Plant Datasets: The datasets of a combined cycle power plant are chosen from UCI Machine Learning Repository to simulate regression transfer learning problem. Totally, 1500 samples, 100 samples, and 300 samples are used as the source training sample, target training sample, and target testing sample, respectively. To generate the discrepancy between the source dataset and the target dataset, the noise N (50, 10) is added to all the target datasets.

3) Generation of QSAR Fish Toxicity Datasets: The datasets of QSAR fish toxicity are selected from UCI Machine Learning Repository. The source training data, target training data, and target testing data have 508 samples, 300 samples, and 100 samples, respectively. The noise N (0.5, 0.3) is added to the output in the target dataset to generate the scene discrepancy.

B. Experimental Setup

The experimental setups including comparison methods, evaluation indices, and other setups can be seen as the follows.

To verify the effectiveness of DK-SOFNN, the proposed DK-SOFNN is compared with three SOFNN, including GM-FTL [20], RS-SOFNN [25], and APSO-FNN [35]. Among them, GM-FTL is the structured transfer learning method, but cannot adjust the structure of the target FNN. RS-SOFNN is the self-organizing FNN without knowledge transfer, and APSO-FNN is the structure and parameter learning method, but neglects the interaction between the structure and parameter and knowledge transfer. The structure of FNN is initialized by experience.

To demonstrate the modeling performance of DK-SOFNN, the root mean square error (RSME), symmetric mean absolute percentage error (sMAPE), and mean absolute square error (MASE) are used to evaluate the regression modeling task. Where MASE can be used for comparison of different data proportions, sMAPE has the preference stability. Three indexes are defined as

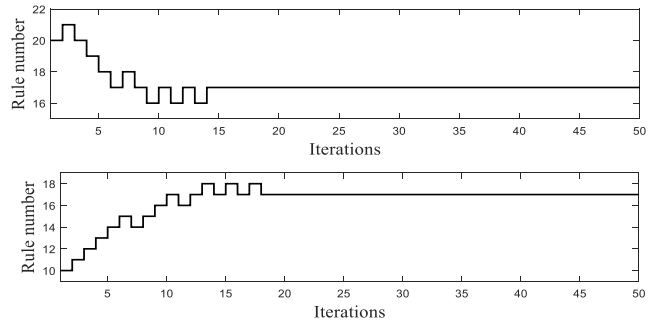

Fig. 4. Structure adjustment process of the source FNN for predicting the Mackey–Glass chaotic system.

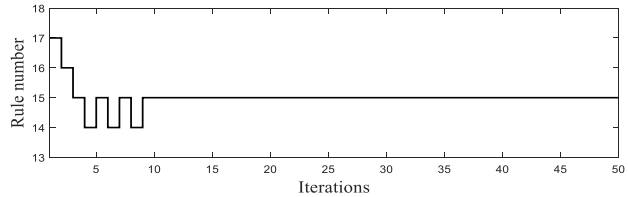

Fig. 5. Structure adjustment process of the target FNN for predicting the Mackey–Glass chaotic system.

$$\text{RMSE} = \sqrt{\frac{1}{N_T}\sum_{t=1}^{N_T}\left(f(\mathbf{x}_T(t)) - y_T(t)\right)^2} \tag{42}$$

$$\text{sMAPE} = \frac{1}{N_T}\sum_{t=1}^{N_T}\frac{2\left|f(\mathbf{x}_T(t)) - y_T(t)\right|}{\left|f(\mathbf{x}_T(t))\right| + \left|y_T(t)\right|} \tag{43}$$

$$\text{MASE} = \frac{1}{N_T}\sum_{t=1}^{N_T}\left(\frac{f(\mathbf{x}_T(t)) - y_T(t)}{y_T(t)}\right)^2. \tag{44}$$

The initial parameters are set as: the weight of the data-driven part is $\alpha^h \in [0.8, 1]$, the weight of the knowledge-driven part is $\beta^h \in [0, 0.2]$, the learning rate of the source scene is $\lambda_S = 0.1$, the learning rate of the target scene is $\lambda_T = 0.01$, the number of iterations is $I = 300$, and the number of fuzzy rules is 20. The mean results of 30 independent runs are shown in Table III with the rule number and the testing RMSE, sMAPE, and MASE.

C. Experimental Results

In this section, comparisons between the proposed DK-SOFNN and other self-organizing FNN methods in terms of generalization performance are presented.

1) Results of Mackey–Glass Chaotic System Prediction: First, the prediction results of the Mackey–Glass chaotic system are shown in Figs. 4–10 and Table III. Fig. 4 is the structural adjustment process of the source FNN under different numbers of starting rules. It can be seen in Fig. 4 that the proposed method can adjust the structure of the source FNN. No matter the initial rule number is set to 10 or 20, the structure of the target FNN is adjusted to 17.

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TABLE III
PERFORMANCE OF DK-SOFNN WITH DIFFERENT METHODS

Datasets	Indexes	**DK-SOFNN**	GM-FTL [20]	RS-SOFNN [25]	APSO-FNN [35]
Mackey–Glass chaotic	Rule numbers	**15**	18	13	12
	RSME	**0.0328**	0.0598	0.0629	0.0778
	sMAPE	**0.0284**	0.0554	0.0582	0.0744
	MASE	**$0.1075e^{-2}$**	$0.3576e^{-2}$	$0.3986e^{-2}$	$0.5225e^{-2}$
combined cycle power plant	Rule numbers	**13**	16	14	9
	RSME	**4.6289**	6.2628	6.4826	6.8165
	sMAPE	**0.0092**	0.0185	0.0189	0.0201
	MASE	**$0.8570e^{-4}$**	$0.1562e^{-3}$	$0.1680e^{-3}$	$0.1858e^{-3}$
QSAR Fish toxicity	Rule numbers	**11**	15	11	10
	RSME	**0.6089**	0.7925	0.8239	0.9021
	sMAPE	**0.0852**	0.1352	0.1402	0.1699
	MASE	**0.0103**	0.0174	0.0189	0.0226

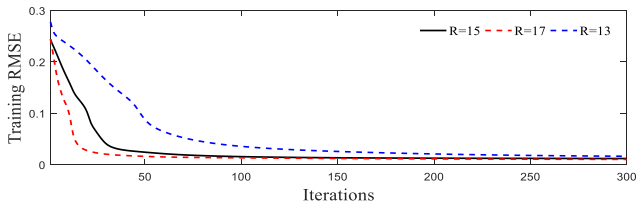

Fig. 6. Comparison of training RMSE with different structures for predicting the Mackey–Glass chaotic system.

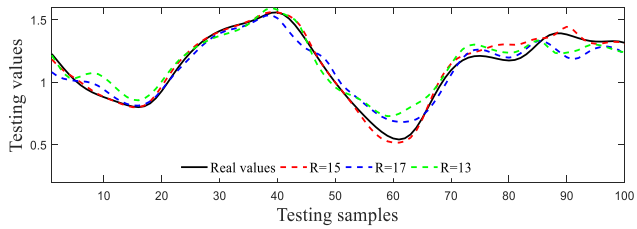

Fig. 7. Comparison of testing values with different structures for predicting the Mackey–Glass chaotic system.

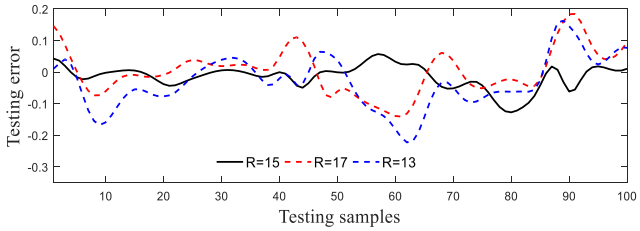

Fig. 8. Comparison of testing error with different structures for predicting the Mackey–Glass chaotic system.

Fig. 5 is the structural adjustment process of the target FNN. From Fig. 5, the initial rule number of the target FNN is set as the rule number of the adjusted source FNN. Then, the structure of the target FNN is adjusted by the structure knowledge of the source FNN and instance of the target FNN. It can be seen that the structure of DK-SOFNN can be adaptively adjusted during the learning process. Due to the low complexity of the target task, useless neurons with high redundancy, low sensitivity, and low contribution are deleted and the structure of target FNN is reduced to improve generalization and reduce computational burden.

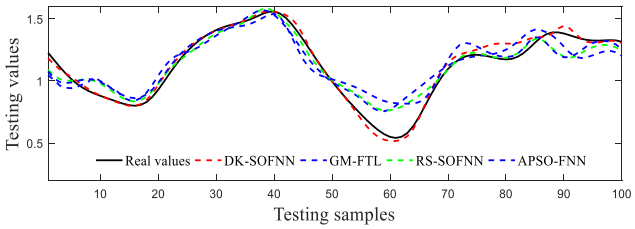

Fig. 9. Comparison of testing values with different methods for predicting the Mackey–Glass chaotic system.

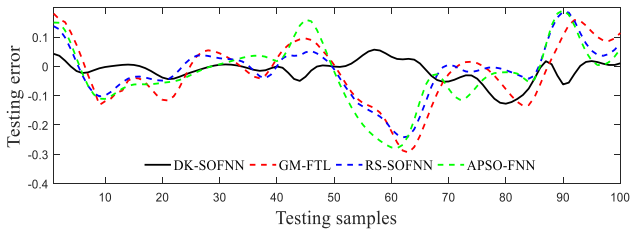

Fig. 10. Comparison of testing error with different methods for predicting the Mackey–Glass chaotic system.

Fig. 6 shows the comparison results of training RMSE with different structures. For the training RMSE from Fig. 6, the rule number of 17 is smaller than that of 15, and the rule number of 13 is larger than that of 15. Figs. 7 and 8 show comparison results of testing values and testing error with different structures. It can be seen from Figs. 7 and 8 that the test results for rules 13 and 17 are worse than those for rule 15.

Fig. 9 shows comparison results of DK-SOFNN with other self-organizing FNNs. Fig. 10 shows the comparison of test errors of DK-SOFNN under other self-organizing FNNs. Figs. 9 and 10 show that the test values of DK-SOFNN can better track the real values, and the test error of DK-SOFNN is the smallest among GM-FTL, RS-SOFNN, and APSO-FNN. Moreover, the detailed results in Table IV illustrate that the proposed KD-SOFNN can obtain the best generalization performance among other methods.

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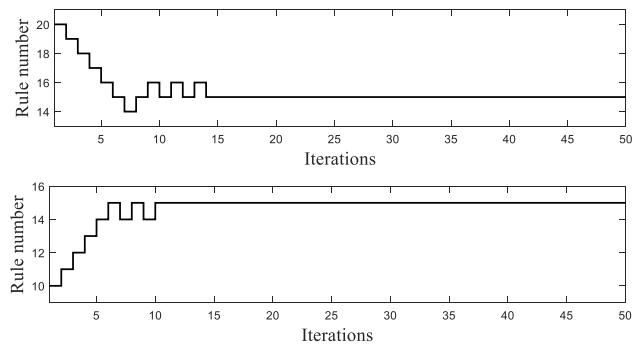

Fig. 11. Structure adjustment process of the source FNN for predicting combined cycle power plant datasets.

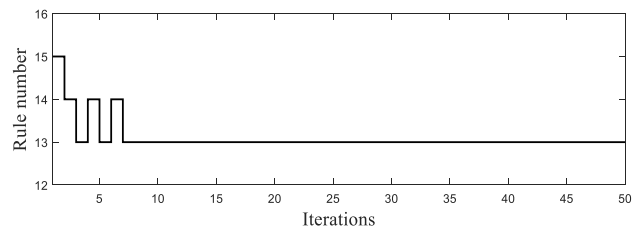

Fig. 12. Structure adjustment process of the target FNN for predicting combined cycle power plant datasets.

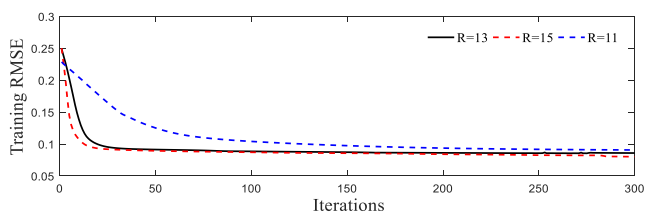

Fig. 13. Comparison of training RMSE with different structures for predicting combined cycle power plant datasets.

2) Results of QSAR Fish Toxicity Prediction: Second, the prediction results of combined cycle power plant are shown in Figs. 11–17 and Table III. Fig. 11 shows the structure adjustment process of the target FNN for predicting combined cycle power plant datasets. It can be seen in Fig. 11 that the proposed method can adjust the structure of the source FNN to 17 under the initial rules of 10 or 20. Fig. 12 gives the structure adjustment process of the target FNN for predicting combined cycle power plant datasets. It can be seen that the structure of DK-SOFNN can be adaptively adjusted during the learning process and the structure of the target FNN is smaller than that of the source FNN.

Fig. 13 shows the comparison of training RMSE with different structures. It can be seen that the larger of structure, the better training accuracy can be acquired. Figs. 14 and 15 are the comparison results of testing values and testing error with different structures. It can be seen from Figs. 14 and 15 that FNN with 13 rules has better prediction accuracy than 15 and 11 rules. Fig. 16 shows the comparison of testing values with different methods for predicting combined cycle power plant datasets. Fig. 17 shows the comparison of testing error with different methods. Figs. 16 and 17 show that the test values of

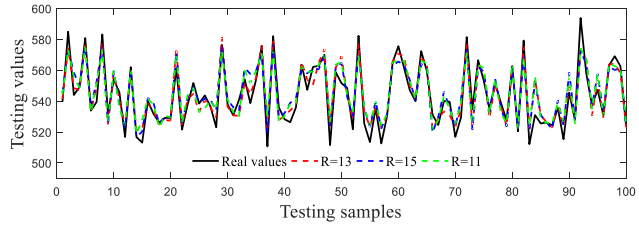

Fig. 14. Comparison of testing values with different structures for predicting combined cycle power plant datasets.

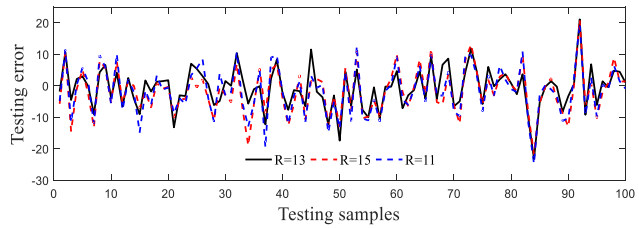

Fig. 15. Comparison of testing error with different structures for predicting combined cycle power plant datasets.

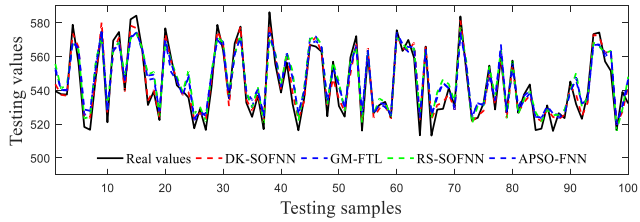

Fig. 16. Comparison of testing values with different methods for predicting combined cycle power plant datasets.

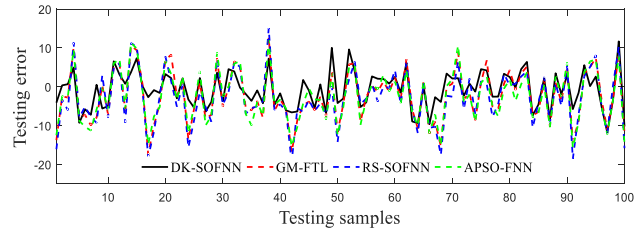

Fig. 17. Comparison of testing error with different methods for predicting combined cycle power plant datasets.

DK-SOFNN can better track the real values, and the test error of DK-SOFNN is the smallest among GM-FTL, RS-SOFNN, and APSO-FNN. The details of comparison results in Table III illustrate that the proposed DK-SOFNN can achieve better generalization performance than other methods.

3) Results of Combined Cycle Power Plant Prediction: Third, the prediction results of QSAR fish toxicity are shown in Figs. 18–24 and Table III. Fig. 18 is the structure adjustment process of target FNN for predicting combined cycle power plant datasets. It can be seen in Fig. 18 that the proposed method can adjust the structure of the source FNN to 11 under the initial rules of 10 or 20. Fig. 19 is the structure adjustment process of target FNN for predicting combined cycle power plant datasets. From Fig. 19, the structure of the target FNN

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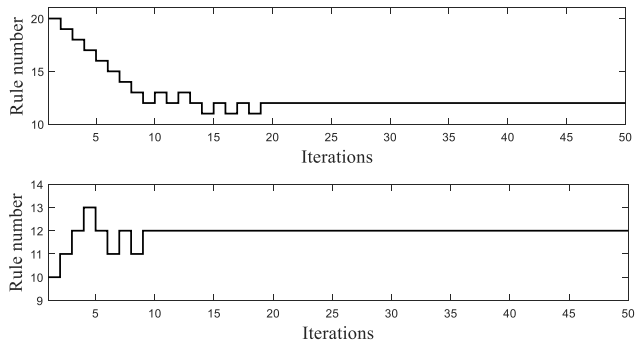

Fig. 18. Structure adjustment process of the source FNN for predicting QSAR fish toxicity datasets.

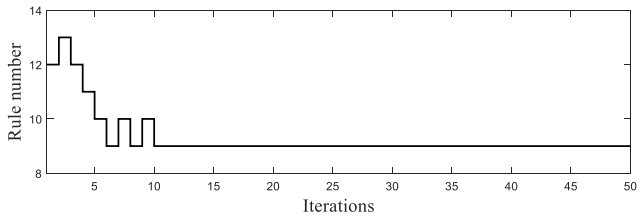

Fig. 19. Structure adjustment process of the target FNN for predicting QSAR fish toxicity datasets.

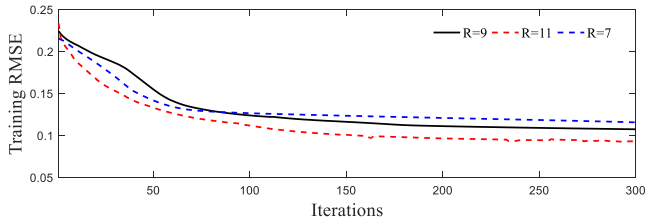

Fig. 20. Comparison of training RMSE with different structures for predicting QSAR fish toxicity datasets.

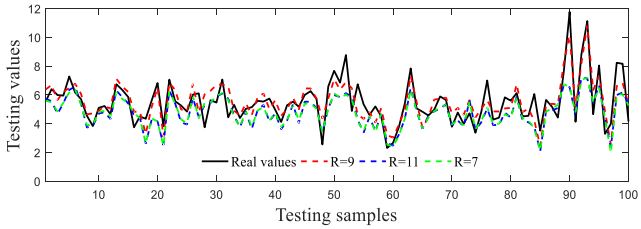

Fig. 21. Comparison of testing values with different structures for predicting QSAR fish toxicity datasets.

is adjusted to 9. Moreover, it can be seen that the structure of the target FNN is smaller than that of the source FNN. Fig. 20 shows the comparison results of training RMSE with different structures for predicting combined cycle power plant datasets. It can be seen that the proposed DK-SOFNN can find a compact and complete structure in the learning process. Moreover, Figs. 21 and 22 are the comparison of testing values and testing error with different structures. The testing results in Figs. 21 and 22 indicate that the proposed DK-SOFNN can predict the combined cycle power plant datasets with higher

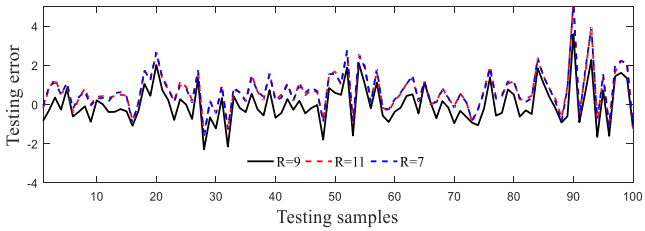

Fig. 22. Comparison of testing error with different structures for predicting QSAR fish toxicity datasets.

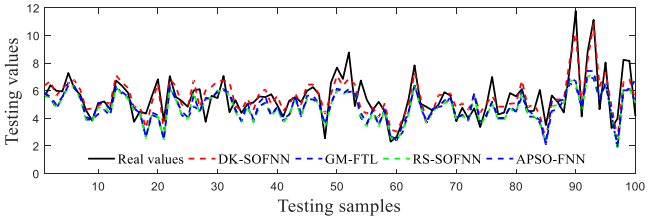

Fig. 23. Comparison of testing values with different methods for predicting QSAR fish toxicity datasets.

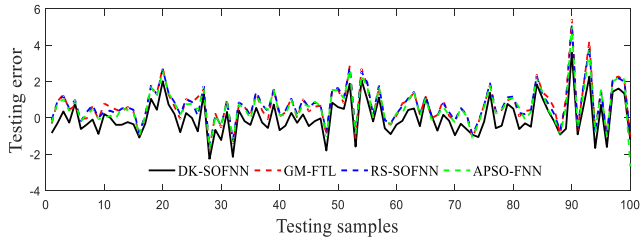

Fig. 24. Comparison of testing error with different methods for predicting QSAR fish toxicity datasets.

accuracy than its adjacent structure, which demonstrate that rule 13 has better generalization.

The testing results of DK-SOFNN and other knowledge and other self-organizing FNN are given in Figs. 23 and 24. The testing results in Fig. 21 illustrate that the proposed DK-SOFNN can approximate the combined cycle power plant datasets with a better generalization performance than GM-FTL, RS-SOFNN, and APSO-FNN. The testing results in Fig. 22 illustrate that the testing values of KDTL-FNN are closer to real values and the testing errors of DK-SOFNN are smaller than GM-FTL, RS-SOFNN, and APSO-FNN. It can be seen from Table III that the proposed DK-SOFNN can obtain a better generalization performance with other methods. The proposed DK-SOFNN has the best performance indexes of RSME, sMAPE, and MASE among the compared methods.

VI. Conclusion

In this article, a DK-SOFNN with the structure compensation strategy and parameter reinforcement mechanism is proposed to improve generalization. Furthermore, the convergence condition of the designed DK-SOFNN is developed based on the Lyapunov stability theory.

In the experiment studies, DK-SOFNN is compared with three related methods: GM-FTL, RS-SOFNN, and APSO-FNN. The synthetic datasets of the Mackey–Glass chaotic

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system have been discussed. Meanwhile, the real datasets of combined cycle power plant and QSAR fish toxicity are used to test the proposed DK-SOFNN. The experimental results show that this proposed DK-SOFNN has an excellent learning ability. The testing results of RMSE, sMAPE, and MASE reveal that this proposed KDTL-FNN is superior to GM-FTL, RS-SOFNN, and APSO-FNN.

The method presented in this article focuses on leveraging the parameter knowledge and structure knowledge to learn FNN. In the future, more diverse knowledge will be mined from the source FNN to optimize the FNN.

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